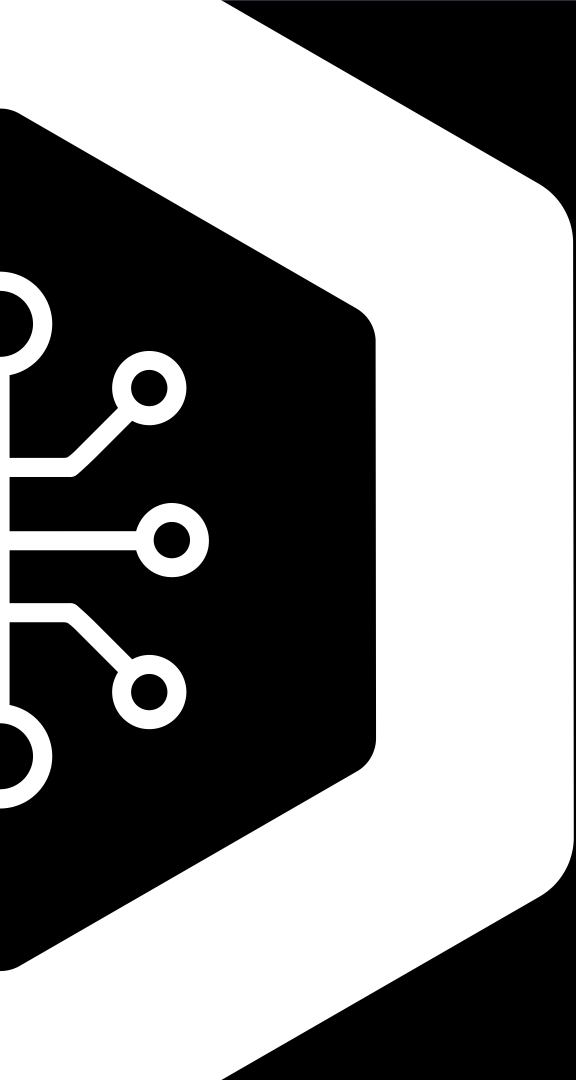




Four-State Logic and Verification Support in Verilator

DAC Open-Source EDA BoF, Long Beach, 2026-07-28

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Progress on all fronts

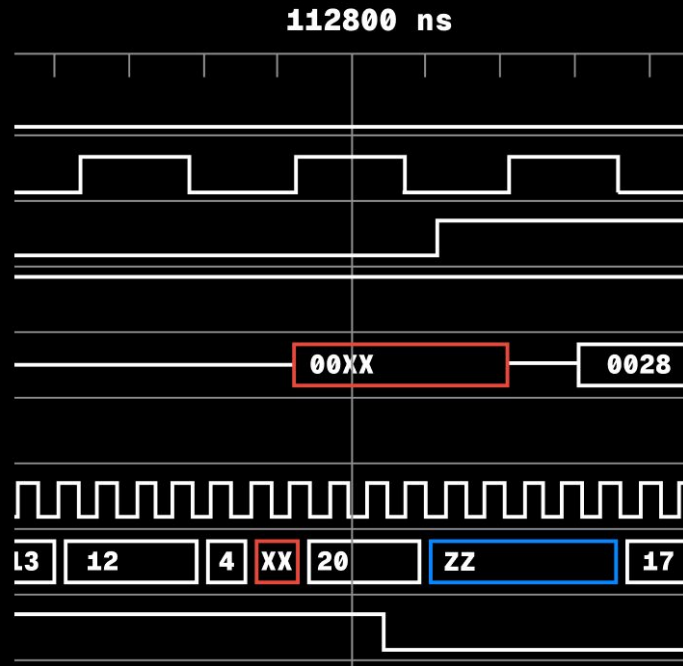
Over 200 PRs by Antmicro in the past year related to:

- Four-state logic
- UVM, classes, scheduling
- Constrained randomization
- Concurrent assertions
- Force/release
- Optimizations



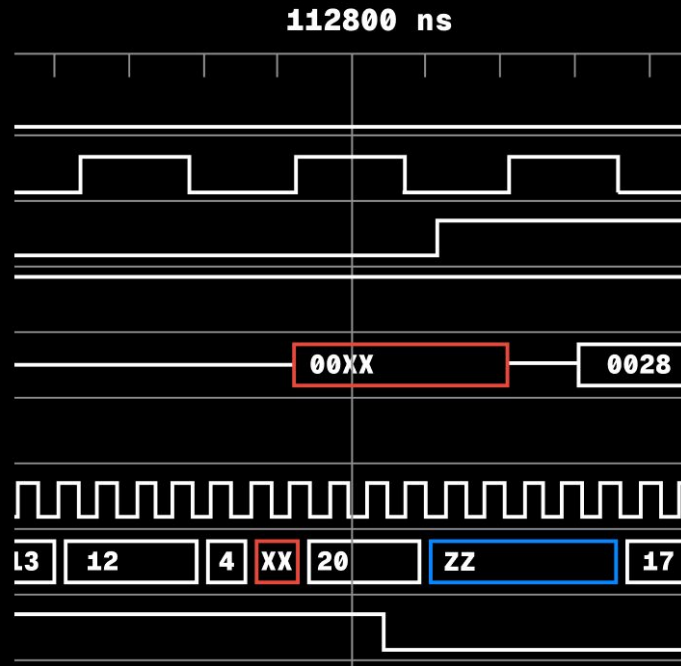
Four-state logic

- Biggest gap in Verilator
- First PRs:
 - github.com/verilator/verilator/pull/7193
 - github.com/verilator/verilator/pull/7437
- Supports:
 - Arithmetic, logic operations on one-dimensional signals of arbitrary length
 - Conditional expressions, short circuiting
 - Edge detection
 - Waveforms



Four-state logic

- Multi-PR effort
- Follow-ups already in progress with support for:
 - Printing x/z
 - Case statements
 - Multidimensional arrays
 - Unpacked types, dynamic arrays
- A development branch can already simulate the CHIPS Alliance I3C core:
github.com/chipsalliance/i3c-core



Interleaved binary and x/z

32 bits

Interpreted as

Four- state variable

Four-state logic

```
logic val1 = 1;
```



```
bit val1_bin = 1;  
bit val1_xz = 0;
```

```
logic val2;
```



```
bit val2_bin = 1;  
bit val2_xz = 1;
```

```
logic val3 = val1 & val2;
```



```
bit val3_bin = (val1_bin | val1_xz)  
               & (val2_bin | val2_xz)  
bit val3_xz = (val1_bin & val2_xz)  
              | (val2_bin & val1_xz)  
              | (val1_xz & val2_xz)
```

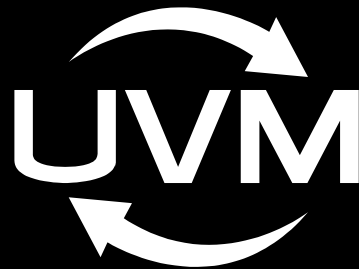
Four-state logic - performance

- Up to 5x longer build time
- 2 to 5x longer run time
- Opt-in via `--fourstate`

Design	Build (2-state)	Build (4-state)	Run (2-state)	Run (4-state)
servant	4.73s	4.84s (1.02x)	0.02s	0.04s (2.0x)
picorv32	4.60s	5.73s (1.25x)	0.09s	0.15s (1.7x)
i3c core	14.26s	64.95s (4.6x)	0.57s	3.04s (5.3x)

UVM support

- UVM testbenches generally compile
 - See github.com/antmicro/verilator-uvm-example
- There are often issues at runtime - but not always!
- Coral NPU is an open source edge AI accelerator by Google
 - Has a UVM testbench that runs binaries and compares against golden model
 - Worked in Verilator out of the box



Coral

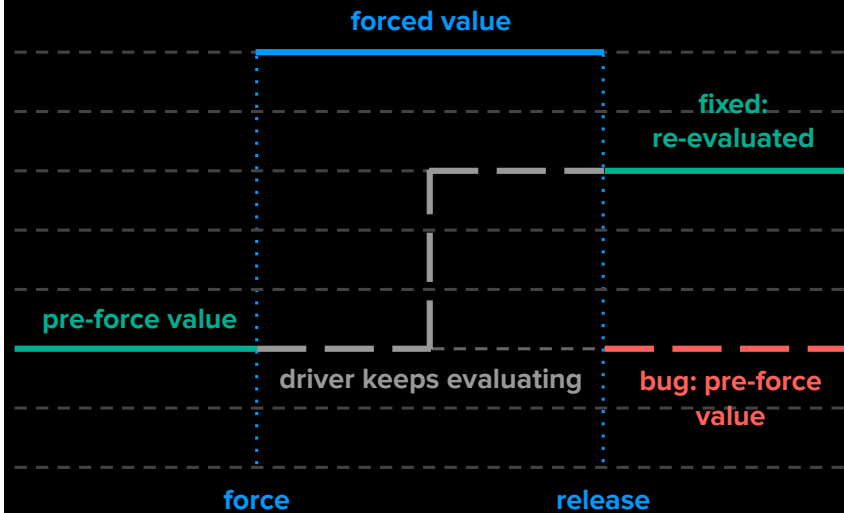
Concurrent assertions

- Not available at all a year ago
- Effort started fall 2025
- Through a community effort, most features are in place
- Remaining work is mostly resolving scheduling issues

```
assert property (  
    @(posedge clk)  
        disable iff (rst) a ##1 b ##2 c)  
    $display("OK");  
else  
    $display("FAIL");
```

IEEE-compliant force/release

- Previously resolved statically at compile time
- It could not handle conditionals, re-evaluation
- Now using a runtime data structure
- Fully IEEE-compliant



SV Tools Project

- github.com/chipsalliance/sv-tools
- Goal is moving the open ecosystem forward
 - UVM support in Verilator
 - Waveform-to-code tracking
 - Coverage gathering and visualization
- TSC meetings every odd week on Fridays at 7:30AM PST





**THANK YOU
FOR YOUR ATTENTION!**

